

Model manifold analysis suggests the human ventral visual system is less like an optimal classifier and more like a feature bank

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Abstract

What do deep neural network (DNN) models actually tell us about the computational principles of visual information-processing in the biological brain? A now common finding in visual neuroscience is that many different kinds of DNNs – with different architectures, tasks, and training diets – are all comparably performant predictors of image-evoked brain activity in the ventral visual cortex. This relative parity of diverse models may at first seem to undermine the common intuition that these models can be used to infer the computational principles that govern the visual brain. In this work, we show to the contrary that comparable brain-predictivity does not preclude the differentiation of these same models in terms of the underlying manifold geometries that define them. To do this, we assess 12 manifold geometry metrics computed on a diverse set of 117 DNN models, curated to include multiple tasks, architectures, and input diets. We then use these metrics to predict how well each model aligns with occipitotemporal cortex (OTC) activity from the human fMRI Natural Scenes Dataset. We find that *manifold signal-to-noise ratio* (a metric previously associated with few-shot learning) is a robust predictor of downstream brain-alignment and supersedes both other manifold geometry metrics (i.e. *manifold capacity*) and downstream task-performance (e.g. top-k recognition accuracy) across multiple image sets (e.g. ImageNet21K versus Places365) and controlled model comparisons (e.g. assessments across ImageNet-1K trained architectural variants only). These results add to a growing body of evidence that the ventral visual stream serves as a basis set (or feature vocabulary) for object recognition rather than as the actual locus of recognition *per se*.

Background & Methods

High-level ventral visual cortex is widely considered the primary substrate of object recognition in the human brain [1; 2; 3], a premise reinforced by the finding that deep neural network (DNN) models trained on image classification are the most predictive models of ventral visual activity to date [4; 5; 6]. A lingering issue, however, is the under-specification of what ‘recognition’ really means in the context of biological brains. In models, these mechanisms are explicitly defined: canonical end-to-end classifiers (e.g. AlexNet [7]) map images onto one-hot category labels; self-supervised models (e.g. SimCLR [8], DINO [9]) learn invariances without

labels; multimodal classifiers (e.g. CLIP [10]) recognize via vision-language similarity scores.

Identifying which motif is most brain-like has been confounded by the fact that many models achieve comparable measures of representational alignment [11] to human visual cortex [6; 12; 13; 14; 15]. One response, inspired by the neuro-connectionist programme [16], has been controlled comparisons that parse model differences by *grouping* them to isolate the influence of training data, architecture, and task [17]. Here, we parse model differences more directly by *sorting them* along scalar axes of representational structure defined by manifold geometry analysis [18; 19; 20], interrogating how this structure relates to object recognition and what models may tell us about its nature in the human ventral visual system. This approach builds on a growing body of work applying manifold geometry to biological and artificial neural representations [21; 22; 23; 24].

Model & metric curation. We curated 12 metrics of manifold geometry from two lines of work: 4 metrics from Chung et al. [18] (derived from replica mean field theory) and 8 from Sorscher et al. [20] (derived from principal components analysis). We evaluated these across 117 DNN models spanning different visual diets, architectures, and tasks [17].

Brain-alignment. Model-to-brain predictivity (henceforth, brain-alignment) was computed as the average voxel-wise encoding score over a 50% train split (N=500) of the ‘Shared NSD1000’ image set [25] using field-standard regularized ridge regression. Models were compared on test-split (N=500) encoding scores for the most brain-predictive layer, selected via nested cross-validation on the train split.

Manifold geometry. Metrics were computed using 50 concept manifolds (50 images $\times$ 50 categories from ImageNet-1K/21K [26; 27] or Places365 [28]), following the protocol of Stephenson et al. [29]. We used the exact source code from the defining works of each metric set.

Results & Discussion

Primary analysis: explaining brain-alignment with manifold geometry. We computed rank-order correlations (ρ_{Spearman}) between brain-predictivity and 12 manifold metrics from N=50 concept manifolds of ImageNet-21K [29]. Eight metrics showed significant correlations ($p < 0.01$): four positive (SNR, capacity, signal,



Correlation of Brain-Predictivity & Manifold Capacity | SNR across Model Sets

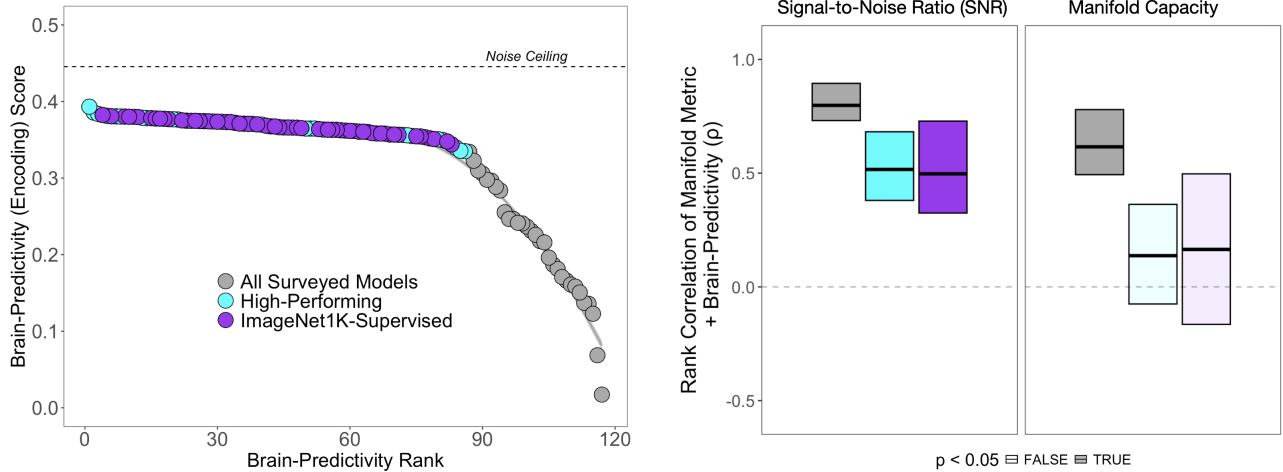


Figure 1: (Left) Brain-predictivity (encoding) scores across 117 models, with model subsets highlighted: all surveyed models (gray), high-performing models above segmented-regression elbow (cyan), and ImageNet-1K-supervised architectural variants only (purple). (Right) Rank correlation (ρ) of manifold SNR and manifold capacity with brain-predictivity across these model subsets. SNR remains a robust, significant predictor of brain-alignment in restricted subsets; capacity does not.

between-concept dimensionality) and four negative (radius, dimensionality, within-concept radius, bias). The strongest predictor was manifold signal-to-noise ratio (SNR) from Sorscher et al. [20], at $\rho = 0.80$, $p < .001$. Manifold capacity [18] was also significant at $\rho = 0.62$, $p < .001$. These results confirm that manifold metrics can be strong predictors of brain-alignment, as previously proposed [20; 30; 31; 32; 33].

Capacity vs. SNR robustness. That *multiple* metrics correlate with brain-alignment raises a question: which geometry is more brain-like? Manifold capacity and SNR instantiate a clear case of this ambiguity. In the extreme, high capacity signals collapse of category information into separable point-estimates – an optimal classifier [18; 19]. Mid-to-high SNR, by contrast, better describes the conditions of few-shot learning [20]. To disambiguate, we tested both in progressively restricted model subsets (Figure 1). In a ‘high-performing’ subset ($N=84$), capacity dropped to non-significance ($\rho = 0.14$, $p = 0.222$) while SNR remained robust ($\rho = 0.52$, $p < .001$). In an architecture-only subset ($N=53$ ImageNet-1K models [34]), we saw even more divergence: capacity $\rho = 0.16$, $p = 0.262$; SNR $\rho = 0.47$, $p < .001$.

Layer-wise analysis and cross-dataset generalization. Further experiments confirmed this divergence even more clearly. In the last hidden layer of category-supervised models – the representation most directly supporting classification – both metrics became strong *negative* predictors of brain-alignment ($\rho = -0.61$ and -0.66), as did classification accuracy itself ($\rho = -0.63$;

c.f. [35]). We additionally recomputed manifold metrics using concept manifolds from ImageNet-1K and Places365. SNR remained a significant positive predictor of brain-alignment in both cases ($\rho = 0.457$, $p < .001$ and $\rho = 0.319$, $p = .020$); capacity was not significantly predictive in either.

General Discussion. What makes for a “good” neural network model of the visual brain? Since the adoption of task-optimized DNNs over a decade ago [36], the dominant answer has been prediction: better models predict brain activity more accurately. For those seeking causal control of biological systems [37; 38], this may suffice. For those seeking understanding, the search remains for variables that supplement prediction with parsimony [16; 33; 39; 40].

Here, we use manifold geometry [19] to link representation to function in models of ventral visual cortex. We find that manifold capacity – scaling with convergence toward separable point-estimates of category identity – is less robust in predicting brain-alignment than manifold SNR, which captures more graded invariance and separability subserving both recognition and few-shot learning. These results add to growing evidence [41; 42; 43; 44; 45; 46; 47] that the ventral visual stream may be less like an optimal classifier and more like a feature bank for whatever compositional process recognition involves downstream. Concurrent work applying manifold geometry directly to brain data and toward manifold-guided neural control [23; 24; 48] offers promising directions for further validation.

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Statements

Use of LLMs: All conceptual development, experiments, analyses, and narrative interpretation of this work were devised, executed, and reviewed by the human authors. LLMs were used as coding assistants (i.e. in the Cursor IDE), for editing the language of the manuscript (i.e. in Overleaf), and were also employed in a custom multi-agent, adversarial "auto-review" pipeline designed by the authors to bring attention to literature or under-score potential errors in the final drafts of the manuscript.